LaborCoin Whitepaper

Abstract

LaborCoin is a decentralized governance system designed to enable collective decision-making and resource allocation for the working class. It combines a non-transferable voting token, DAO-based governance, and controlled execution through dedicated smart contracts to create a transparent, accountable, and scalable coordination mechanism.

The system enforces a one-person, one-vote model and restricts execution to predefined actions, minimizing attack surface while preserving flexibility through governance.

In addition, LaborCoin integrates a bonding curve exchange mechanism that enables continuous liquidity, deterministic pricing, and treasury accumulation to support striking workers and collective action.

1. Introduction

Modern economic systems concentrate decision-making power in centralized institutions, limiting the ability of workers to coordinate and act collectively. While blockchain technology enables decentralized coordination, many implementations prioritize speculation over governance.

LaborCoin addresses this gap by providing:

A governance-first architecture
Equalized voting power
Transparent, on-chain decision execution
Controlled and auditable action pathways
A treasury-building mechanism tied to real economic participation

The goal is not to create a speculative asset, but a functional coordination system capable of sustaining worker-led movements.

2. System Overview

LABR (primary token)
LABRV (voting token)
DAO governance (Aragon)
Executor contracts
Bonding curve exchange

Participants → LABR → LABRV → Proposals → DAO → Executors → Treasury → Worker Support

This architecture ensures decisions are enforced through constrained execution while maintaining economic backing through the bonding curve.

3. Governance Model

Voting Mechanism

1 LABRV per participant
Non-transferable (soulbound)
Equal voting weight

Prevents capital-based governance capture and ensures democratic participation.

Thresholds

25% participation
67% approval

High thresholds ensure legitimacy and prevent minority control.

4. Token Design

Contract Address:
0x460DD873A1D2a41e77410B125cD3027C5FEd2f78

Total Supply: 1,000,000,000
Initial Supply: 100,000,000

Emission Model

50,000,000 LABR released per tranche
1:1 release ratio up to 50% of total supply
1:0.8 release ratio beyond 50%

This structure encourages early participation while gradually introducing scarcity.

Constraints

Transaction Limit: 5,000 LABR per transaction
Wallet Limit: 10,000 LABR per wallet
Cooldown: 12 hours between transactions

These constraints are enforced at the contract level to prevent accumulation of disproportionate influence and reduce volatility.

5. Bonding Curve Exchange

Exchange Contract:
0xD0692ec758bb852421B702B187b6439f74f8Bf3b

Pricing Model

Quadratic bonding curve with smooth price progression
Price increases as total tokens sold increases
Approximate range: $1 → ~$15

Economic Behavior

Early participants enter at lower prices, incentivizing initial adoption.
As participation grows, price increases smoothly rather than abruptly.
This discourages speculation-driven spikes and instead rewards sustained participation.

Treasury Model

All purchases deposit POL into the contract treasury
Sales withdraw from treasury liquidity
10% of each purchase is routed to the DAO treasury

This creates a direct link between participation and available support funds.

6. Execution Architecture

DAO → Executor → Contract

Executors

PauseExecutor → controls trading state
TreasuryExecutor → distributes funds
MintExecutor → mints voting tokens

Execution is constrained to prevent arbitrary control and reduce attack surface.

7. Deployment

Network: Polygon

LaborCoin (LABR):
0x460DD873A1D2a41e77410B125cD3027C5FEd2f78

LaborCoin Exchange:
0xD0692ec758bb852421B702B187b6439f74f8Bf3b

LaborCoin DAO:
0x0C2e5679153593b82a84eAB5CA90895BB291Cec4

DAO Registration (V2)
0xFFc3499A71b806C3919f4B54D236b151cFdCB453

LaborVote (LABRV v6):
0x113579220515cd59b884Ea2379b4C369025246e2

DAO Governance (V2):
0x3edfa4BE01e7a244357198260f2432094690aB65

Pause Executor (V2):
0xe2a791FD057bBA5AEa0efcfb8b97f9de74cC7c06

Treasury Executor (V3):
0x260e5c1dA01a51A3059FDb2163914AC9ff83c0F7

8. Security Model

Restricted execution pathways
Immutable economic parameters
Non-transferable governance
High consensus thresholds

Additional Protections

Oracle-based pricing validation
Slippage protection
Circuit breaker (pause mechanism)
Reentrancy protection

9. System Invariants

The following conditions are designed to always hold:

No participant can obtain more than one voting token
Governance decisions cannot execute arbitrary contract logic

Treasury withdrawals cannot exceed available balance
Token supply cannot exceed maximum defined limits
Pricing remains deterministic and independent of external markets

These invariants define the core guarantees of the system.

10. Threat Model

LaborCoin assumes adversarial behavior as a baseline condition.

Adversarial Actors

Wealthy participants attempting to accumulate influence
Coordinated voting blocs attempting governance capture
Smart contract exploiters targeting liquidity
External actors manipulating price inputs

Attack Surfaces

Bonding curve pricing
Treasury liquidity
DAO participation thresholds
Executor permissions
Oracle feeds

11. Economic Attack Vectors

Whale Accumulation

Mitigation: Wallet caps, transaction limits, and non-transferable voting tokens.

Liquidity Drain Attacks

Mitigation: Increasing price curve, cooldown enforcement, treasury constraints.

Speculative Volatility

Mitigation: Deterministic pricing independent of external markets.

12. Governance Risks

Low Participation

Mitigation: High quorum requirements.

Coordination Attacks

Mitigation: 67% approval threshold.

Voter Apathy

Acknowledged as a social limitation inherent to democratic systems.

13. Failure Scenarios

Oracle Failure

If price oracle data becomes invalid, transactions relying on price inputs may fail or pause.

Treasury Depletion

If treasury liquidity is insufficient, sell transactions may fail until replenished.

Low Participation

Governance may stall if participation thresholds are not met.

Contract Pause

The system may enter a paused state under emergency conditions to prevent further damage.

14. System Assumptions

Participants engage in governance
Worker solidarity can be coordinated
Blockchain infrastructure remains secure
Oracle pricing remains reliable

LaborCoin minimizes trust in individuals but depends on collective participation.

15. Use Cases

Strike funding
Mutual aid
Collective bargaining
Resource allocation

16. Conclusion

LaborCoin shifts blockchain systems from speculation to coordination infrastructure. It combines governance, economics, and execution into a unified system capable of supporting real-world collective action.